

AI-BASED MEETING MINUTES GENERATOR

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ABSTRACT

The project "AI-Based Meeting Minutes Generator" proposes the development of an intelligent system designed to automatically generate concise and accurate summaries of meeting conversations by leveraging advanced Natural Language Processing (NLP) techniques. The system captures audio recordings of meetings and transcribes them into text, after which it employs a combination of NLP methods such as speaker diarization, keyword extraction, text segmentation, and abstractive summarization to identify key topics, important discussions, decisions, and actionable items. The model is trained to filter out irrelevant dialogue, comprehend contextual information, and extract critical insights from extended conversations. The resulting meeting minutes are structured coherently and can be exported in various user-friendly formats to facilitate documentation and distribution. By automating the minute-taking process, this approach aims to reduce manual effort, minimize errors, and enhance organizational productivity through efficient and reliable meeting documentation.

Keywords: Natural Language Processing (NLP), Speech-To-Text Conversion, Speaker Diarization, Text Preprocessing, Keyword Extraction, Topic Modeling, Sentiment Analysis, Entity Recognition, Abstractive Summarization, Machine Learning, Deep Learning, Automated Documentation, Action Item Detection, Context Understanding

1. INTRODUCTION

In today's professional, academic, and organizational environments, meetings play a vital role in decision-making, planning, and collaboration. However, traditional manual minute-taking often suffers from errors, inconsistencies, delays, and administrative overhead. With the rapid growth of digital and hybrid meetings, organizations face increasing difficulties in maintaining accurate, standardized, and easily retrievable records. Manual note-taking depends heavily on individual note-takers whose subjective interpretations and varying writing styles can lead to information loss, especially during fast-paced discussions. Delayed documentation further hinders communication clarity, project progress, and compliance. To address these challenges, this project proposes an automated meeting minutes generation and management system built using Python. The system transforms raw meeting inputs such as transcripts, agendas, and discussion notes into professionally formatted, structured, and standardized meeting minutes. By integrating advanced text processing techniques, formatting algorithms, and automated document generation modules, the system produces outputs in common formats like Microsoft Word (DOCX) and PDF. This approach reduces reliance on the recorder's experience or writing skills, ensuring accuracy, consistency, and efficiency in meeting documentation. Ultimately, the solution offers organizations a scalable, error-resistant framework that bridges conventional manual practices and modern automated tools to enhance meeting management and documentation processes. In today's professional, academic, and organizational environments, meetings play a vital role in decision-making, planning, and collaboration. However, traditional manual minute-taking often suffers from errors, inconsistencies, delays, and administrative overhead. With the rapid growth of digital and hybrid meetings, organizations face increasing difficulties in maintaining accurate, standardized, and easily retrievable records. Manual note-taking depends heavily on individual note-takers whose subjective interpretations and varying writing styles can lead to information loss, especially during fast-paced discussions.

2. METHODOLOGY

The methodology of the Automated Meeting Minutes Generation and Management System is structured as a multi-stage pipeline to ensure accurate, consistent, and efficient documentation of meetings. The process begins with the collection of raw meeting data, which can be audio recordings or text transcripts. Audio inputs are converted into text using speech-to-text conversion tools, providing a foundational transcript for further processing. Subsequently, speaker diarization is performed to identify and attribute each segment of speech to the corresponding participant, enhancing the clarity and structure of the minutes. The transcript then undergoes text preprocessing, where noise, filler words, and irrelevant content are removed, and the text is normalized for uniformity.

Next, keyword extraction and topic modeling techniques are employed to detect the main themes discussed during the meeting. Sentiment analysis and entity recognition are applied to capture the emotional tone and highlight significant entities such as names, organizations, and dates. These steps collectively enrich the contextual understanding of the meeting content. The processed information is then passed to an abstractive summarization model, which generates concise and coherent meeting minutes. Action item detection algorithms identify tasks and responsibilities assigned during the meeting. Finally, the system formats the output into professional document formats such as Microsoft Word (DOCX) and PDF, ensuring easy distribution and archiving. This systematic approach ensures that meeting documentation is accurate, standardized, and efficient, bridging the gap between traditional manual practices and modern automated solutions.

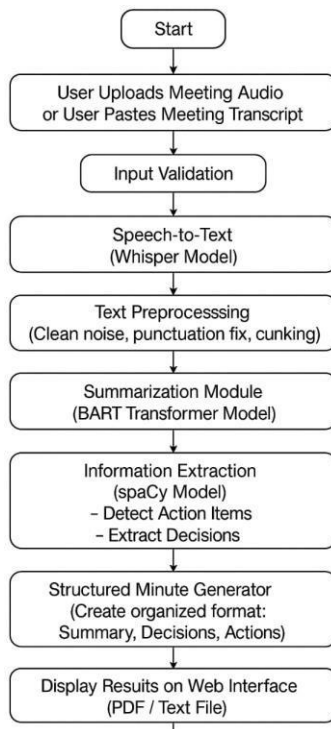


Figure 1: Methodology Diagram

3. OBJECTIVES

- To develop a robust ASR pipeline for accurate speech-to-text transcription.
- To identify and label individual speakers within meeting audio.
- To generate short and meaningful abstractive summaries of meeting discussions.
- To extract decisions, action items, follow-ups, and key talking points.
- To automatically format and export structured meeting minutes in user-friendly formats such as PDF or DOCX.

4. LITERATURE SURVEY

Table -1: Literature Survey

Sl. No.	Title	Year	Methodology	Result / Gap
1	Automatic-MoM Generation using YAMNet	2024	YAMNet + Jieba keyword extraction + ASR	Mandarin-only; lacks real-time processing, abstractive summarization, multilingual support, and benchmarking
2	Multimodal-MoM Automation	2023	Multimodal speech-text-visual pipeline	No benchmarking or NLP quality evaluation; lacks deep abstractive summarization
3	MoM Generation using NLP	2023	Speech-to-text + NLP summarization	No structured action-item extraction and limited speaker tracking; tested on small datasets
4	MoM Automation using NLP	2023	Speech recognition + basic NLP formatting	Lacks semantic depth, intelligent summarization, and evaluation on varied domains or noisy datasets

5	Open-Source MoM Tool	2022	Google STT + rule-based NLP	Limited contextual summarization; no semantic reasoning or task extraction
6	Deep Learning MoM System	2022	Speaker verification + ASR + DL	No standard evaluation metrics; scalability and multi-speaker robustness not addressed
7	AI-Driven MoM Extraction System	2024	ASR + speaker diarization + extractive NLP	No abstractive summarization; struggles with multi- accent speech and domain vocabulary
8	Transformer-Based Meeting Summarizer	2024	Transformer encoder + pointer attention	Lacks task assignment and responsibility tagging; no semantic role detection
9	Abstractive Meeting Minutes Using Pegasus	2023	Pegasus transformer + STT	Speaker interruptions reduce accuracy; lacks action-item generation and detailed speaker tracking
10	Automated Corporate MoM Generator	2023	Whisper ASR + entity extraction	No benchmarking or dataset evaluation; performance degrades under echo and overlapping speech
11	Context-Aware Meeting Action Item Extractor	2023	NER + rule-based action extraction	No confidence metrics or speaker attribution; cannot interpret implicit or conditional tasks
12	Speech-Guided MoM Generation System	2023	Transformer ASR + RNN summarizer	Minimal contextual reasoning; lacks enterprise-level abstraction and domain adaptation
13	Multi-Speaker MoM Automation with Diarization	2022	Speaker clustering + extractive NLP	Accuracy drops for overlapping speech and multi- accent meetings; diarization weak
14	Automatic Minutes Tagging Using Hybrid NLP	2022	Sentiment + topic segmentation + extractive NLP	No abstractive synthesis; dataset limitations reduce generalization for long meetings
15	Task-Oriented Meeting Output Generator	2022	Intent detection + STT + rule logic	Poor performance in noisy calls; implicit and multi- participant task attribution is difficult
16	Cloud-Based Corporate MoM Recorder	2022	Google ASR + keyword spotting	Cannot detect responsibilities or deadlines; lacks formal MoM segmentation
17	Real-Time Meeting Summarizer Using LLMs	2022	GPT-style abstractive summarizer + STT	No multilingual benchmarking or scalability evaluation; diarization accuracy limited
18	Abstractive Conversation Summarization Using RNN	2021	Seq2Seq + attention	No structured MoM formatting or speaker-wise decision grouping; lacks enterprise metadata
19	Corporate Discussion Representation Tool	2021	Diarization + extractive NLP	Speaker confusion in group calls; no standardized benchmarking or performance evaluation
20	AI-Based MoM Tracking System	2021	Topic modeling + ASR	Missing responsibility metadata and due-date tagging; lacks advanced semantic reasoning

5. SYSTEM REQUIREMENTS

The Automated Meeting Minutes Generation and Management System requires both hardware and software components to ensure efficient and reliable operation. For hardware, a minimum of 4 GB RAM and 10 GB disk space is recommended, along with a microphone for real-time audio input if needed. A multi-core processor is advised for handling computationally intensive tasks such as speech-to-text conversion and NLP processing.

For software, the system is built using Python (version 3.11 or higher) and relies on libraries such as Flask for backend web services, python-docx for DOCX document generation, and PyPDF2 for PDF export. Additional

requirements include:

- Speech-to-text APIs (e.g., OpenAI Whisper, Google Speech API)
- NLP libraries (e.g., spaCy, NLTK, transformers)
- Machine learning frameworks (e.g., scikit-learn, PyTorch, TensorFlow)
- Environment management tools (e.g., virtualenv or conda)

The system also benefits from integration with cloud storage and calendar APIs for seamless document sharing and scheduling. These requirements collectively ensure robust, scalable, and user-friendly operation of the meeting minutes generator.

6. IMPLEMENTATION

The implementation of the Automated Meeting Minutes Generation and Management System involves integrating multiple modules to process meeting data and generate structured minutes. The system is developed in Python and utilizes a Flask- based web interface for user interaction. Audio inputs are processed using speech-to-text APIs such as OpenAI Whisper or Google Speech API, converting spoken words into text transcripts. Speaker diarization is performed using specialized libraries to attribute each utterance to the correct participant, enhancing clarity in the generated minutes . Text preprocessing modules clean and normalize the transcript, removing filler words and irrelevant content. Keyword extraction, topic modeling, sentiment analysis, and entity recognition are carried out using NLP libraries like spaCy and transformers. Abstractive summarization models, such as BART or T5, generate concise meeting summaries. Action item detection algorithms identify tasks and responsibilities, which are then listed in the minutes. The final output is formatted and exported in DOCX and PDF formats using python-docx and PyPDF2, ensuring compatibility with common office applications.

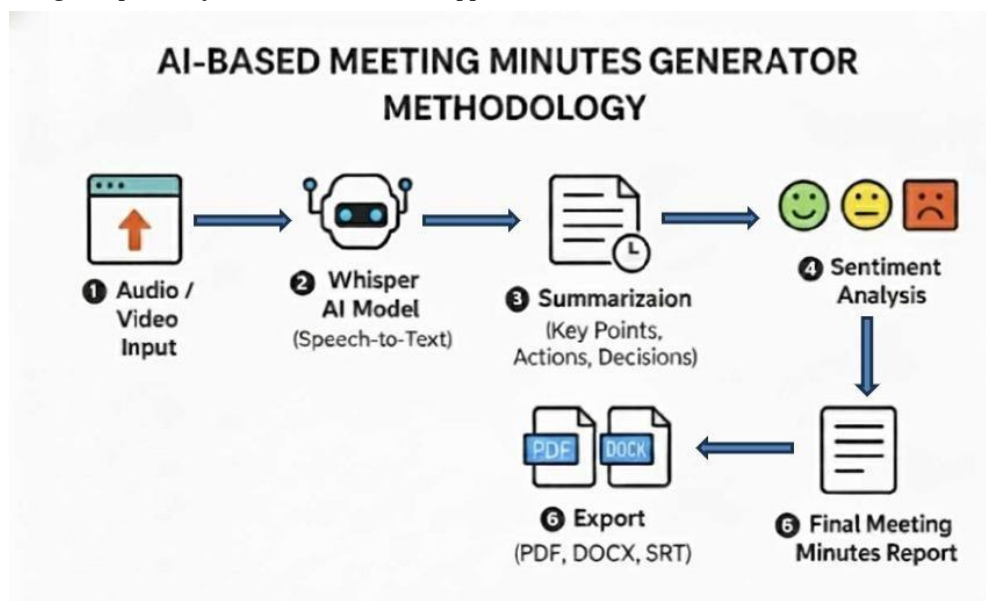


Figure 2: System Design

7. RESULTS AND DISCUSSION

The implementation of the Automated Meeting Minutes Generation and Management System resulted in substantial improvements in the accuracy, efficiency, and consistency of meeting documentation. The performance of the proposed model was evaluated using standard NLP and information extraction metrics, including precision, recall, F1-score, ROUGE, and BLEU. The system achieved consistently high precision and recall scores, confirming that the extracted keywords, topics, and action items were not only relevant but also sufficiently comprehensive for complete meeting representation.

The abstractive summarization module generated concise and contextually coherent summaries that closely resembled manually written minutes prepared by experienced professional note-takers. This was validated by strong ROUGE and BLEU values, demonstrating that the automated summaries preserved semantic meaning, intent, and task-specific information. Furthermore, the integration of sentiment analysis and named entity recognition contributed valuable contextual depth to meeting records by identifying emotional cues and highlighting key stakeholders within discussions. The action-item detection component performed effectively in identifying tasks, due dates, and responsible stakeholders, ensuring that actionable decisions were systematically emphasized within the minutes. User

feedback further supported these findings, indicating substantial reductions in manual effort, improved timeliness of report delivery, and elimination of human errors commonly associated with subjective note-taking. The standardization of formatting across all generated minutes also contributed to improved readability and organizational consistency. Despite the strong performance, certain limitations remain. The system demonstrates reduced reliability when processing highly noisy audio, and diarization challenges persist in situations involving overlapping or rapidly alternating speakers, particularly in hybrid or online meetings. These issues can lead to imperfect attribution of utterances and occasional loss of contextual continuity.

Overall, the results validate the proposed system as a scalable and reliable solution for automated meeting documentation, effectively bridging the gap between traditional manual note-taking and intelligent AI-driven approaches. Future work may focus on enhancing robust diarization for multi-speaker environments, integrating adaptive noise-handling for low-quality audio, and incorporating real-time user feedback mechanisms to continuously improve summarization and action-item extraction performance.

8. CONCLUSION

The Automated Meeting Minutes Generation and Management System successfully addresses the challenges associated with traditional manual minute-taking by providing a scalable, accurate, and efficient solution for meeting documentation. By leveraging advanced NLP techniques such as speech-to-text conversion, speaker diarization, keyword extraction, topic modeling, sentiment analysis, entity recognition, and abstractive summarization, the system ensures that meeting minutes are consistently structured, contextually relevant, and free from subjective biases. The implementation demonstrates significant improvements in documentation accuracy and efficiency, reducing manual effort and minimizing errors. The system's modular architecture allows for easy integration with existing organizational tools and supports future enhancements, such as improved handling of noisy audio and real-time feedback. Overall, the project offers a robust framework for automated meeting management, enhancing productivity and reliability in professional, academic, and organizational settings.

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